

DAVID ABABIO AWUNI

Independent Researcher | Language-Model Evaluation & Multilingual NLP

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I study how evaluator choice and prompt wording affect conclusions about language models. My recent work examines family-conditioned preferences in LLM judges and answer switching around African-language content, with an emphasis on controlled experiments and reproducible evaluation.

EDUCATION

BSc Computer Science, First Class Honours

Jan 2021–Aug 2024 · Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

Thesis: A Multi-Layered Hybrid Deep Learning Framework for Cyber-Physical Intrusion Detection in Climate-Monitoring IoT Systems. Supervisor: Dr. Linda Banning.

RESEARCH MANUSCRIPTS

Awuni, D. A., Achenie, L. E. K., Partey, B. T., Owusu, E. G., & Sowah, N.-N. D. (2026). Who Judges Matters: Measuring Family-Conditioned Preference in LLM-as-Judge Panels. arXiv:2609.17857 [cs.CL]. [\[arXiv\]](#) [\[Code\]](#)

Awuni, D. A., Gyening, R.-M. O. M., & Owusu, E. G. (2026). AfriSyCo: Measuring Assertive Framing, Verification, and Wording Sensitivity Around African-Language Content. arXiv:2609.17853 [cs.CL]. [\[arXiv\]](#) [\[Code\]](#)

CONFERENCE PRESENTATION

Awuni, D. A., Gyening, R.-M. O. M., & Twum, F. (2024). *Enhancing Student Engagement in Online Lecture Streaming Platforms through Real-Time Quizzes*. E-Learning International Conference, KNUST, Kumasi, Ghana. Presenter: David Ababio Awuni.

RESEARCH EXPERIENCE

Independent Researcher

2026–Present · Ghana

Who Judges Matters

- Built a pairwise-evaluation pipeline across Llama, Qwen, Gemma, and Yi, with 194 prompts and 9,312 judgments, counterbalancing response order.
- The study introduced a corrected family-preference estimator that holds candidate family fixed; the evaluated panel showed a 6.7-percentage-point same-family lift (95% CI: 5.3–8.4).
- Assessed robustness using independent human-consensus labels, statistical sensitivity analyses, and a separate float16 judging replication.

AfriSyCo

- Evaluated seven open-weight checkpoints on 100 source questions in Hausa, Igbo, Kinyarwanda, Swahili, Twi, and Yoruba, analysing 1,415 model–language–item observations with correct initial answers.
- Used native-language follow-ups and a separate 2×2 experiment crossing assertion and verification; the controlled experiment kept questions and targets in African languages and follow-up framing in English.
- In the controlled experiment, assertive framing increased target selection by 30.4 percentage points (95% CI: 28.4–32.3), averaged over three prompt families; effects varied by wording and checkpoint.

Undergraduate Researcher

Jan–Aug 2024 · KNUST

- Developed a multi-layer intrusion-detection framework combining environmental telemetry and network traffic; evaluated sensor-level and CNN-BiLSTM classifiers on ToN_IoT and CICIoT2023.
- Implemented binary detection and 25-class attack classification, using SHAP to examine feature importance and interpret model decisions.

TEACHING AND MENTORING

Research and Teaching Assistant

Oct 2024–Aug 2025 · Department of Computer Science, KNUST, Kumasi, Ghana

- Supported laboratory instruction and assessment for 150+ undergraduates in Computer Security and Data Communications, including programming guidance and troubleshooting.
- Mentored 20+ undergraduates on thesis design, research methods, implementation, and presentation; provided feedback on graduate research drafts to clarify methods and align conclusions with evidence.

ADDITIONAL TECHNICAL EXPERIENCE

Edustream

Jan–Sep 2024 · Student research and software project; KNUST, Kumasi, Ghana

- Co-developed desktop, web, and mobile components for live lectures and student interaction; adapted an NLP question-generation pipeline to create contextual quizzes from lecture content.

Frontend Developer Intern

Aug–Dec 2023 · Codextra, Accra, Ghana

- Developed web and mobile features, reviewed pull requests, and presented weekly technical progress reports across concurrent projects.

HONOURS AND AWARDS

Vice-Chancellor's Students' Excellence Award, KNUST

2024

First Runner-Up, All-Rounded Exemplary Student.

Fifty-50 Club Scholarship

2021–2024

Merit- and need-based undergraduate scholarship.

METHODS AND TOOLS

Programming and analysis: Python, SQL, pandas, NumPy; JavaScript and Dart.

Machine learning: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, PEFT/LoRA.

Evaluation and statistics: LLM-as-judge evaluation, multilingual experiments, human annotation, inter-annotator agreement, bootstrap inference, permutation tests, GLMs, GEE, and Bradley–Terry models.

Research tools: Git, Linux, Docker, AWS, Google Colab, LaTeX.

Languages: Twi/Akan (native); English (fluent).

SELECTED LEADERSHIP AND SERVICE

Senate Speaker

Aug 2023–Aug 2024 · Science Students' Association, KNUST

- Presided over 24 Senate sessions with 50+ student representatives and led review of an annual association budget exceeding GHS 100,000.

Organiser and Lead Facilitator

Oct–Nov 2023 · Programming Boot Camp

- Designed and delivered practical training in Python, web development, and digital literacy for 80+ students in Kumasi.

Academic Committee Head

Aug 2022–Sep 2023 · Computer Science Society, KNUST

- Coordinated peer tutoring and coached departmental quiz teams, including a third-place finish in the Prof. Amekudzi College Quiz.